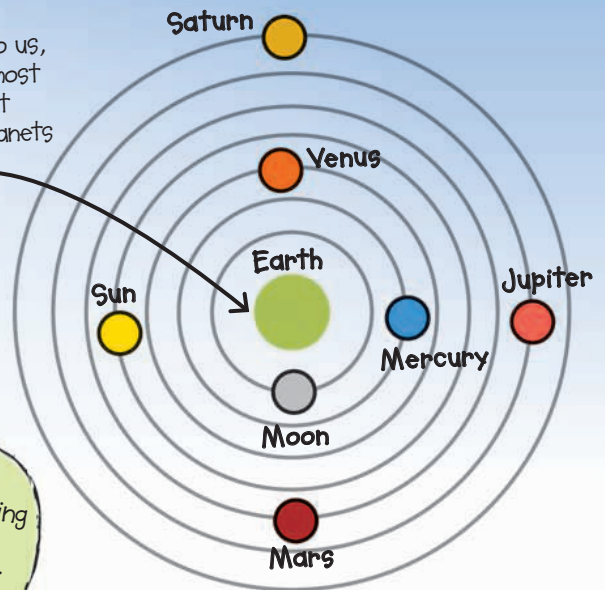


Galileo versus God

In one of his books, Galileo explained how the Earth orbits the Sun. The church didn't like this because it taught that the Earth is the **center of the Universe**. In 1616, Galileo was accused of *heresy* and forbidden from teaching or talking about his theories. He didn't obey the church's request for long, and, in 1633, he was sentenced to **LIFE IN PRISON** unless he renounced his beliefs.

It might seem silly to us, but before Galileo most people believed that the Sun and the planets orbited the Earth.



Did you know?
Galileo didn't get everything right. He thought that the tides were caused by oceans sloshing about as Earth orbits the Sun.

Falling for gravity

Galileo performed lots of different **experiments**. He proved that heavy and light objects **FALL AT THE SAME SPEED** and how gravity makes falling objects *accelerate*. He also worked out that the only thing that affects how fast a pendulum swings (like the ones you find in some old clocks) is its length and the strength of gravity.

Galileo dropped two different cannonballs from the Tower of Pisa to prove that objects fall at the same speed.



Trailblazer
Albert Einstein called Galileo the "father of modern science." His discoveries undermined the power of the church and helped give scientists the freedom to talk about their work without fear of persecution.



NICOLAS COPERNICUS (1473–1543) used **MATHEMATICAL MODELS** to prove that the Earth orbits the Sun.



The nobleman **TYCHO BRAHE** (1546–1601) made the most accurate **ASTRONOMICAL MEASUREMENTS** of his day.